Model Development Phase Template

Date	25 June 2026
Team ID	SWTID-2026-5589
Project Title	Credit Card Approval Prediction
Maximum Marks	4 Marks

Initial Model Training Code, Model Validation and Evaluation Report

The initial model training code is showcased through a screenshot. The model validation and evaluation report includes classification reports, accuracy, and confusion matrices for multiple models, presented through respective screenshots.

Initial Model Training Code:

Logistic Regression Model

```
def logistic_reg(xtrain, xtest, ytrain, ytest):
    lr = LogisticRegression(solver='liblinear')
    lr.fit(xtrain, ytrain)
    ypred = lr.predict(xtest)
    print("LogisticRegression")
    print('Confusion matrix')
    print(confusion_matrix(ytest, ypred))
    print('Classification report')
    print(classification_report(ytest, ypred))
    return lr
```

Random Forest Classifier

```
def random_forest(xtrain, xtest, ytrain, ytest):
    rf = RandomForestClassifier()
    rf.fit(xtrain, ytrain)
    ypred = rf.predict(xtest)
    print("RandomForestClassifier")
    print('Confusion matrix')
    print(confusion_matrix(ytest, ypred))
    print('Classification report')
    print(classification_report(ytest, ypred))
```

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XG Boost Model

```
def g_boosting(xtrain, xtest, ytrain, ytest):
    gb = GradientBoostingClassifier()
    gb.fit(xtrain, ytrain)
    ypred = gb.predict(xtest)
    print("GradientBoostingClassifier")
    print('Confusion matrix')
    print(confusion_matrix(ytest, ypred))
    print('Classification report')
    print(classification_report(ytest, ypred))
```

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Decision Tree Model

```
def d_tree(xtrain, xtest, ytrain, ytest):
    dt = DecisionTreeClassifier()
    dt.fit(xtrain, ytrain)
    ypred = dt.predict(xtest)
    print("DecisionTreeClassifier")
    print('Confusion matrix')
    print(confusion_matrix(ytest, ypred))
    print('Classification report')
    print(classification_report(ytest, ypred))
```

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Model Validation and Evaluation Report:

<table>
<tr><th>Model</th><th>Classification Report</th><th>Accuracy</th><th>Confusion Matrix</th></tr>
<tr><td>Logistic Regression</td><td>Classification report
precision recall f1-score support
0 1.00 0.99 0.99 1037
1 0.99 0.99 0.99 905
accuracy 0.99 1942
macro avg 0.99 0.99 0.99 1942
weighted avg 0.99 0.99 0.99 1942</td><td>99%</td><td>LogisticRegression
Confusion matrix
[[1025 12]
[5 900]]</td></tr>
<tr><td>Random Forest Classifier</td><td>Classification report
precision recall f1-score support
0 0.99 0.99 0.99 1037
1 0.98 0.99 0.99 905
accuracy 0.99 1942
macro avg 0.99 0.99 0.99 1942
weighted avg 0.99 0.99 0.99 1942</td><td>98%</td><td>RandomForestClassifier
Confusion matrix
[[1022 15]
[8 897]]</td></tr>
<tr><td>Gradient Boosting Classifier</td><td>Classification report
precision recall f1-score support
0 1.00 1.00 1.00 1037
1 1.00 1.00 1.00 905
accuracy 1.00 1942
macro avg 1.00 1.00 1.00 1942
weighted avg 1.00 1.00 1.00 1942</td><td>100%</td><td>GradientBoostingClassifier
Confusion matrix
[[1036 1]
[4 901]]</td></tr>
</table>

Decision Tree Classifier	Classification report precision recall f1-score support 0 1.00 1.00 1.00 1037 1 1.00 1.00 1.00 905 accuracy 1.00 1942 macro avg 1.00 1.00 1.00 1942 weighted avg 1.00 1.00 1.00 1942	100%	DecisionTreeClassifier Confusion matrix [[1035 2] [1 904]]